Guangxing Hu

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EDUCATION

North Carolina State University Raleigh, NC
- Ph.D. in Computer Science (Advisor: Prof. Frank Mueller) 08/2023 – Present
- CGPA: 4.00 / 4.3
- *Relevant Courses*: CSC501 Operating Systems; CSC791 Efficient Deep Learning; CSC505 Design and Analysis of Algorithms; CSC548 Parallel Systems

Huazhong University of Science and Technology Wuhan, China
- M.Sc. in Computer System Architecture 09/2019 – 06/2023
  - Thesis: Research on Performance Anomaly Detection and Root Cause Analysis for Storage Systems in Data Centers
  - Advisor: Prof. Ke Zhou
  - Focus: Data Center, Storage System, Anomaly Detection, Root Cause Analysis, Machine Learning.

Hainan University Haikou, China
- B.Eng. in Computer Science and Technology 09/2015 – 06/2019
- CGPA: 3.78/4.0 Rank: 3/125

EXPERIENCE

Oak Ridge National Laboratory Oak Ridge, TN
Graduate Research Internship 06/2024 – 08/2024
- Conducted advanced research on integrating Persistent Memory (PM) technologies within high-performance computing systems, developing and evaluating benchmarks under diverse workloads and operational conditions.
- Collaborated with experts to design and implement innovative file systems that leverage the unique characteristics of PM, optimizing deep learning training using novel storage technologies.

NCSU Systems Lab, North Carolina State University Raleigh, NC
Graduate Research Assistant 08/2023 – Present
- Analyzed various file systems and systematically examined the unique characteristics and performance metrics of NFS, SSD, fsdax, and Persistent Memory (PM) technologies. Designed benchmarks that integrated multiple methods, including mmap, fio, and memkind, to assess the access speeds of each system under various workloads and operational conditions.
- Utilizing techniques such as LD_PRELOAD, implemented I/O redirection for PM operations, allowing for the dynamic interception and redefinition of standard library calls without modifying existing application code.

North Carolina State University Raleigh, NC
Graduate Teaching Assistant 08/2023 – 12/2023
- CSC246 Concepts and Facilities of Operating Systems for Computer Scientists (S'23)

Wuhan National Laboratory for Optoelectronics Wuhan, China
Master's Student/Research Assistant 09/2019 – 06/2023
- Analyzed the anomaly detection research of time-series data on data centers for storage systems using machine and deep learning methods, studied the anomaly detection method based on the Generative Adversarial Network in-depth, and analyzed the root cause of system failure by random walk and other methods.
- Contributed to constructing simulation infrastructure in the data center and proposed a framework for anomaly detection to detect data and status anomalies by combining Vector Quantized Variational AutoEncoder (VQ-VAE) and Shapelet.
- Conducted analysis on how visual chirality affects the performance of image hashing and performed experiments on supervised, self-supervised and unsupervised hash approaches.
- Implemented a disk failure prediction system for integration of high precision and low false alarms based on SMART data for heterogeneous disks by utilizing a Long Short-Term Memory (LSTM) based Siamese Network.

PUBLICATIONS

Google Scholar Profile: https://scholar.google.com/citations?user=j8Yq5W8AAAAJ&hl=en

* means equal contribution

Poster

[1] Guangxing Hu, Awais Khan, Frank Mueller. "A Zero-Copy Storage with Metadata-Driven File Management Using Persistent Memory." ***SC***, 2024.

Publications

[1] Tianming Jiang, Yunchuan Guan, Ke Zhou, Guangxing Hu, Ji Zhang, Wei Fang, Zhuo Cheng, Ping Huang. "SPAE: Lifelong Disk Failure Prediction via End-to-End GAN-based Anomaly Detection with Ensemble Update." ***Future Generation Computer Systems***, 2023.

[2] Yanzhao Xie*, Guangxing Hu*, Yu Liu, Zhiqiu Lin, Ke Zhou and Yuhong Zhao. "How Visual Chirality Affects the Performance of Image Hashing." ***Neural Computing and Applications***, 2022.

[3] Yangtao Wang, Yanzhao Xie, Lisheng Fan, Guangxing Hu. "STMG: Swin transformer for multi-label image recognition with graph convolution network." ***Neural Computing and Applications***, 2021.

AWARDS & SCHOLARSHIPS

- National Grand Prize of Massive Storage Algorithm Competition. 2023
  This prestigious award recognizes the most effective algorithm designed to optimize cloud storage on a large scale.
- Academic Scholarship of Huazhong University of Science and Technology. 2019 - 2022
  This scholarship honors academic excellence at Huazhong University of Science and Technology.
- Grand Prize, Group Programming Ladder Tournament, China Collegiate Computing Contest. 2018
  This award is given to the group that demonstrates superior coding proficiency and strategic thinking in the China Collegiate Computing Contest.
- The Principal Scholarships of Hainan University (Top 2%). 2018
  An accolade for academic distinction, this scholarship is bestowed upon students who rank in the top 2% of their class.
- The Most Innovative and Outstanding Student Award of Hainan University. 2018
  This award is given to a student who has shown remarkable innovation and overall excellence in their academic and extracurricular endeavors.
- Merit Student Award of Hainan University. 2018
  The Merit Student Award is presented to students with exceptional character and academic achievements.
- The Second Scholarships of Hainan University 2016, 2017
  This scholarship acknowledges students who have achieved high academic performance and who have made significant contributions to university life.

SKILLS

Programming Languages

- Familiar (≥ 4 years of experience): Python, C, C++
- Intermediate ($1 \sim 3$ years): MATLAB, Shell, SQL, Java
- Basic (≤ 1 year): HTML, TOML, OpenMP, Cuda

Technologies

- **Databases:** MySQL (2 years)
- **File and Storage:** Linux VFS (1 year), HDFS (1 year)
- **Operating Systems:** Linux (5 years)
- **Big Data:** Hadoop (1 year), Spark (< 1 year)
- **Tools**: VSCode (4 years), Pycharm (4 years), Git (3 years), Vim (3 years), LaTeX (3 years), Scikit-learn (4 years), Pytorch (4 years), Vdbench (1 year), Fio (1 year), CMake (1 year), GDB (1 year)

Human Languages

- Chinese (Native), English (Fluent)

PROJECTS

IO Redirection for Persistent Memory

- Use the characteristics of persistent memory to redirect IO streams to achieve rational storage and fast access to data.

Autopilot

- Designed an end-to-end architecture encompassing data collection, analysis, processing, model training, and anomaly detection.

Disk Failure Prediction in Cloud

- Utilized a pair of Siamese Long Short-Term Memory (LSTM) neural networks to intricately analyze Self-Monitoring, Analysis, and Reporting Technology (SMART) data to detect and predict potential faults in hard disk drives in the cloud environment.